

COMPUTATIONAL RECORD FOR REVISION R32

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1. REPRODUCTION AND IMMUTABLE INPUTS

The scientific source is in `revisions/2026-09-24-r32/replication`. The exact maintenance model is defined in `coupled.py`; the registered proposals and frozen-parameter records are generated by `study.py`; the interval-wide scalar proof is checked by `boundary.py`; an independently organized checker and malformed-input tests are in `audit.py`. Tables and a file-hash manifest are generated by `report.py`. All rational calculations use Python's arbitrary-precision fractions. Neural fitting uses CPU binary64 PyTorch. The recorded initial environment used Python 3.13, NumPy 2.3.5, and PyTorch 2.10.0+cpu. Every execution records its actual Python and platform rather than implying hardware-independent timing.

From the repository root, a complete fresh reproduction is:

```
python -m pip install numpy==2.3.5
python -m pip install torch==2.10.0 \
    --index-url https://download.pytorch.org/whl/cpu
R=revisions/2026-09-24-r32
python "$R/replication/study.py" "$R/results" --horizon 4
python "$R/replication/study.py" "$R/results" --horizon 8
python "$R/replication/study.py" "$R/results" --horizon 12
python "$R/replication/boundary.py" \
    --output "$R/results/boundary.json"
python "$R/replication/audit.py" --folder "$R/results"
python "$R/replication/report.py"
```

Do not use Python's `-O` flag: the exact construction deliberately refuses execution when assertions are disabled. The independent checker uses explicit exceptions for its substantive comparisons. A failed run is retained as an unresolved record, not silently omitted from a table. The final report generator requires the complete 42-case cohort and all certificates before producing publication tables.

The five publication entry points are `ECTA_R32.tex`, `SUPP_R32.tex`, `RESPONSE_R32.tex`, `COMPUTATION_R32.tex`, and `HISTORY_R32.tex`. The first four use the repository's unmodified Econometric Society class. Build the article before the supplement, whose external references point to the article. The history PDF includes the five already compiled R30 submission objects without modifying their pages. The build script performs multiple LaTeX passes and records actual exit status. A source manifest hashes the replication, records, and paper sources; the final delivery manifest separately hashes the PDFs.

2. WHAT THE COSTS INCLUDE

Each neural proposal is fitted in a fresh subprocess. Its elapsed time includes interpreter/framework initialization and coefficient serialization; coefficient loading is charged at compilation. Subsequent costs are separated into witness construction, own-policy evaluation, eligible-set construction, dynamic minimum-cost computation, final evaluation, internal verification, and serialization. The independent final checker is timed separately and added in the

main timing table. Reference solves, pointwise comparisons, sensitivity-envelope generation, and supplementary audit experiments are reported separately, not silently charged as necessary optimizer work.

The class-based pipeline is executed even when the installed proposal eventually qualifies for zero intervention, so that conservative over-revision is visible. The independent audit additionally evaluates the incumbent and writes a distinct zero-intervention certificate when the upper-witness test passes. The incremental own-evaluation and zero-certificate checking time is recorded. A production implementation can perform this test before cost optimization; the paper does not retroactively replace its measured class-pipeline timing by that hypothetical shortcut.

Times are single executions on the recorded machine. The same witness is reused across proposals in the study, but its full measured construction cost is charged to each cold solution for a standalone accounting. The exact dynamic-programming reference and the certified greedy baseline use the same primitives; they optimize operating return rather than the added revision-cost objective. Library memory is included in process peak resident memory. Portable uncompressed certificate bytes and mathematical piece/bit counts are reported separately.

3. FAILURE RETENTION AND IMPLEMENTATION CORRECTIONS

The initial shared-process timing cohort and its source remain under `results/attempt`s/`phase-timing-v0`. The final shape-preserving encoder corrects a general issue with independent ordinate rounding; the old tested inequalities are not relabeled as failed, and the old source is not removed. A batch horizon-12 execution interrupted by the local execution limit is retained as an interrupted attempt. It is distinct from a model or certificate failure. The final horizon runs are complete.

No seed, horizon, tolerance, or adverse comparison was dropped. The predeclared stress rule is not included in neural performance counts. The zero-cost corollary was checked for every case, not only the spline successes. The main pipeline and independent checker record 42 primary passes, the additional check records 12 zero-intervention passes, and all eight malformed-certificate mutations are rejected. The rational scalar proof is a separate calculation; its inherited endpoint payoff provenance remains explicit.

4. AUDIT TRAIL AND LIMITATIONS OF VERIFICATION

The checker recomputes whole-cell inequalities using exact affine overlays and point values. It does not invoke the maximizing/minimizing envelope routine or train a model. It shares representation and primitive affine operations with construction, so it is not a fully independent software stack or a machine-checked proof assistant. The supplementary mathematical proofs identify the properties these operations must satisfy. Tiny exhaustive scenario-tree checks and malformed certificates supplement, but do not replace, that proof.

The scientific guarantee is for exact compiled deployment. No hardware floating-point neural execution is covered. Economic sensitivity is over named computed policies with normalized implementation units. The original stopped-economy full-domain target and the neural-specific speedup question remain separate from the certified coupled computation. Those boundaries are part of the record and are not converted into experimental successes by a successful build.